MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)

THIRDSEMESTEREXAMINATION

(2024ADMISSIONONWARDS)

MG3DSCIAM200 EmbeddedSystemProgrammingTools

Duration:1.5hours **MaximumMarks:50**

**Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest(I)and Appreciation (Ap)*

Part A

MultipleChoiceQuestions

Answer **All** Questions

Eachquestioncarries1mark

1	Whichofthefollowingis*not*avalidfundamentaldatatypeinPython? a)int b)list c) float d)str	U	1
2	WhichoperatorisusedinPythontocomparetwovaluesforequality? a)= b):= c)== d)===	U	1
3	WhichcomponentinRaspberryPi4isusedtoconnectexternaldisplays? a)HDMIport b)GPIO pins c)USB3.0 d)CSI port	U	1
4	Whichofthefollowing methodsisusedtoconvertallcharacters ina string touppercasein Python? a)string.uppercase() b) string.UPPER() c)string.upper() d) string.toUpper()	U	2
5	WhatisthecorrectwaytoinitializeGPIOpinsinRaspberryPi usingtheRPi.GPIOlibrary? a)GPIO.setmode(GPIO.OUTPUT) b) GPIO.setmode(OUTPUT) c)GPIO.setmode(GPIO.BOARD) d) GPIO.init(GPIO.BOARD)	U	2
6	WhichPythoncontrolstructureisbestsuitedforexecuting a blockofcode repeatedlywhen aconditionistrue? a)if b) for c)while d) switch	U	2
7	WhichPythonmoduleismost commonlyusedforcreating GUI applications? a)pygame	U	3

	b) tkinter c) matplotlib d) pandas		
8	InGUIprogramming,whichprogrammingparadigmbestdescribeshow the application flow is controlled? a)Procedural programming b)Event-drivenprogramming c)Batchprocessing d)Streamprocessing	U	3
9	WhichgeometrymanagerinTkinterplaceswidgetsinrowsandcolumns? a)pack b)grid c)place d)align	U	3
10	WhichofthefollowingisusedtocustomizethetextcolorofaTkinter widget? a)fg b)bg c)textcolor d)color	U	3
			1 * 10 = 10

Part B

ShortAnswer Questions

Answerany**4**Questions

Eachquestioncarries5marks

11	ExplainthestructureofaPythonprogram withasuitableexample.	U	1
12	Differentiatebetweenlist,tuple,andrange withexamples.	U	1
13	Explainthedifferencebetweenif,if-else,for,andwhilestatements in Python with examples.	U	2
14	DescribeatleastfourbasicstringoperationsinPythonwith examples	U	2
15	WriteaPythonprogramusingTkintertocreateabasicwindowwith a label and a button.	U	3
16	Explaintheuseanddifferencebetweenpack(),grid(),andplace() geometry managers in Tkinter. Provide examples.	U	3
			4 * 5 = 20

Part C

Essay

QuestionsAnswerany**2**Q

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Eachquestioncarries10marks

17	Describethedifferent typesofoperatorsinPythonwith examples.	U	1
18	Explainthesyntaxrules,conventions,andstructureofaPython program.	U	1
19	ExplainGPIOprogramminginRaspberryPi.Includedetailsonpin numberingschemes,initializingI/Opins,importinglibraries,andthe useofdigitalread/writefunctionswithcode examples.	U	2
20	ExplaintheprocessofcreatingabasicGUIapplicationusingTkinter in Python.	U	3
			2 * 10 = 20